

```
void setup()
{
  Serial.begin(115200);
  // Initialise wifi connection
  wifiConnected = connectWifi();

  if(wifiConnected) {

    // Define your devices here.
    espalexa.addDevice("Light 1", firstLightChanged); //simplest definition, default state off
    espalexa.addDevice("Light 2", secondLightChanged, 255); //third parameter is beginning state (here fully on)
    espalexa.addDevice("Light 4", forthLightChanged, 255); //third parameter is beginning state (here fully on)
    espalexa.addDevice("Light 5", fifthLightChanged, 255); //third parameter is beginning state (here fully on)
    espalexa.addDevice("Light 6", sixthLightChanged, 255); //third parameter is beginning state (here fully on)
    espalexa.addDevice("Light 7", seventhLightChanged, 255); //third parameter is beginning state (here fully on)

    device3 = new EspalexaDevice("Light 3", thirdLightChanged); //you can also create the Device objects yourself like here
    espalexa.addDevice(device3); //and then add them
    device3->setValue(128); //this allows you to e.g. update their state value at any time!

    espalexa.begin();

  } else
  {
  }
```

Fig. 3: Code setup function

```
void secondLightChanged(uint8_t brightness) {
    Serial.println("off");
    //do what you need to do here
}

void thirdLightChanged(uint8_t brightness) {
    Serial.println("Discovery");

    //do what you need to do here
}

void forthLightChanged(uint8_t brightness) {
    Serial.println("Volumeup");
    //do what you need to do here
}

void fifthLightChanged(uint8_t brightness) {
    Serial.println("Starplus");
    //do what you need to do here
}

void sixthLightChanged(uint8_t brightness) {
    Serial.println("Volumedown");
    //do what you need to do here
}

void seventhLightChanged(uint8_t brightness) {
    Serial.println("Next");
    //do what you need to do here
}

// connect to wifi - returns true if successful or false if not
```

Fig. 4: ESP code consisting of commands to send from serial to Arduino

and then add the required device(s) in the code so that Alexa receives the function of individual buttons of the TV remote.

Note. Since espalexa cannot be made to work as a remote device, use the following hack instead. Create a function for light, which will trigger the IR LED of the TV and enable you to control it like an IR remote. For example, for the ‘on’ button, create a function named as Light 1 in the code. Continue adding 13 to 14 separate functions, say, Light 1 up to Light 13 (as per requirement) for performing other operations.

In the setup function, set a serial baud rate of either 9600 or 115200 so that commands can be successfully passed on to Arduino.

Create a loop function for setting the function of devices such as Light 1, Light 2, etc. Whenever Alexa is given a command, it will trigger functions Light 1, Light 2, etc associated with a specific connected device, that is, TV remote. These functions correspond to commands like volume up/down, TV on/off, next channel, favourite channel, etc.

Arduino code. Write the code for Arduino to retrieve commands from espalexa and then trigger the IR LED to perform as a TV remote control.

Include the Infrared library and paste the remote button array in the code (see figure below). Then create a setup function to set the serial port using `Serial.begin (baud rate)`. Enter a specific baud rate (preferably 115200).

```
#include <irUtils.h>  
  
#include <IrSenderPwm.h>  
  
static const frequency_t necFrequency = 38400U;  
static const unsigned long BAUD = 115200U;  
static const pin_t PIN = 3U;  
  
// NEC(l) 122 29 with no repetition: powers on many Yamaha receivers  
static const microseconds_t array0[] = {  
    417U, 278U, 167U, 278U, 167U, 611U, 167U, 444U, 167U, 611U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U,  
};  
  
// NEC(l) 122 29 with no repetition: powers on many Yamaha receivers  
static const microseconds_t array1[] = {  
    417U, 278U, 167U, 278U, 167U, 611U, 167U, 444U, 167U, 611U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U,  
};  
  
static const microseconds_t array2[] = {  
    417U, 278U, 167U, 278U, 167U, 611U, 167U, 444U, 167U, 611U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U,  
};  
  
// NEC(l) 122 29 with no repetition: powers on many Yamaha receivers  
static const microseconds_t array3[] = {  
    417U, 278U, 167U, 278U, 167U, 611U, 167U, 444U, 167U, 611U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U,  
};  
  
// NEC(l) 122 29 with no repetition: powers on many Yamaha receivers  
static const microseconds_t array4[] = {  
    417U, 278U, 167U, 278U, 167U, 611U, 167U, 444U, 167U, 611U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U, 167U, 278U,  
};  
  
// NEC(l) 122 29 with no repetition: powers on many Yamaha receivers  
static const microseconds_t array5[] = {
```

Fig. 5: Arduino code